POLARIS

Spring Research Project 2024

Mentor: William Koll

Mentee: --

Project Title: Quasiparticle interference mapping of Cu(111) surface

Background: Electrons in materials occupy states within energy bands. The material's electronic properties are determined by the degree to which these bands are filled. Whereas completely full or empty bands lead to insulating behavior, metals (such as Cu) have a partially filled band which enables bulk electrical conductivity. The Fermi surface is a conceptual boundary inside the partially filled band that separates occupied from unoccupied electron states. Although the Fermi surface "lives" in momentum space, its structure can be deduced from real-space data thanks to Fourier analysis. In this project, scanning tunneling microscopy (STM) images of the [111] facet of a Cu crystal surface will be analyzed using Fourier techniques. Due to the peculiar shape of the Cu Fermi surface, the electron momentum is confined to lie in-plane, resulting in a 2D electron gas (2DEG). Electrons in this gas scatter off defects atop the surface, which modifies the local density of electronic states. Due to the wavelike properties of quantum mechanical particles, electrons of opposite momentum interfere constructively/destructively to produce a standing wave pattern with an apparent wavelength that depends on the energy of the tunneling electrons. By analyzing these standing waves, one can extract the effective mass of electrons in the surface state.

Goals:

1. Plot STM data using either Python or other plotting software of their choice, and apply filters as necessary
2. Measure wavelength of standing wave pattern using Fourier analysis
3. Plot measured wavelength vs sample bias
4. Extract effective mass of the surface electrons via fit of data to theory

---

POLARIS

Project Calendar:

Dates:	Tasks:
Week 1 (2/12) Intro to Project	Reading to understand Fourier transform. Readings to understand basic condensed matter concepts (band structure, Fermi surface)
Week 2 (2/19) Outside of Class Work	Download files Try to plot things in Python or using plotting software like Gywddion Continue readings if necessary
Week 3 (2/26) In class work day	Confirm that data is being plotted correctly Fourier transform data Perform basic measurements such as linecut
Week 4 (3/4) Outside of Class Work	Readings about theoretical prediction of standing wave pattern vs sample bias Implement basic image filtering techniques to improve measurements.
Spring Break	None
Week 5 (3/18)	Settle on a strategy or strategies for measuring standing pattern wavelength(s). Can be done with linecuts in real image or FFT, but student should at least compare with FFT. Student can start recording wavelength vs sample bias measurements.
Week 6 (3/25) In Class Work Day	Finish measurements of standing wave pattern wavelength. Plot results.
Week 7 (4/1) In Class Work Day	Fit parabola to above results From fit, extract effective mass
Week 8 (4/8) Poster Work Day	Work on poster
Week 9 (4/15) Present!	

Reading List:

- Basic Fourier transform intro:
  - 3B1B video: (https://www.youtube.com/watch?v=spUNpyF58BY&ab_channel=3Blue1Brown)
- Some basic STM resources:
  - Youtube video: (https://www.youtube.com/watch?v=HE2yE8SvHmA&ab_channel=LaPhysiqueAutrement)
  - Textbook by Chen (for reference only – do NOT need to read whole thing)
- Some basic band structure resource
  - Khan academy youtube video: (https://www.youtube.com/watch?v=3npADYVtQOM&ab_channel=KhanAcademyIndia-English)
  - Max Lein lecture (https://www.youtube.com/watch?v=1ZpFq0W067o&t=706s&ab_channel=MaxLein)
  - CH2 of “Condensed Matter Essentials”
- Previous STM studies of Cu(111)

Resource List:

- Gupta group lab (0105)
  - NOT necessary, but optionally available
- Free, open-source scanning probe microscopy plotting software like
  - Gywddion (http://gwyddion.net/)
  - SpectraFox (https://www.spectrafox.com/)
- Python + IDE of choice